Cambridge (CIE) IGCSE Chemistry
Paper 3: Theory (Set E)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

/ 80

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Materials

- A copy of the Periodic Table of elements is required for this paper.

Scan here to mark your mock exam
or visit the mock exams landing page for this course

1 (a) Define the term *isotope*.

....

....

(2 marks)

(b) The table gives information about four particles, **A**, **B**, **C** and **D**.

Complete the table.

The first line has been done for you.

particle	number of protons	number of electrons	number of neutrons	nucleon number	symbol or formula
A	6	6	6	12	C
B	11	10	12		
C	8		8		O^{2-}
D		10		28	Al^{3+}

....

....

....

....

....

....

(7 marks)

2 (a) Choose a gas from the following list to answer the questions below. Each gas may be used once, more than once or not at all.

ammonia   argon   chlorine   ethene   hydrogen

carbon dioxide   nitrogen   carbon monoxide   oxygen

Which gas

i) Is a noble gas?

[1]

ii) Is an acidic oxide?

[1]

iii) Can be polymerised?

[1]

iv) Is the active component of air?

[1]

v) Is used in the treatment of water?

[1]

....

....

....

....

....

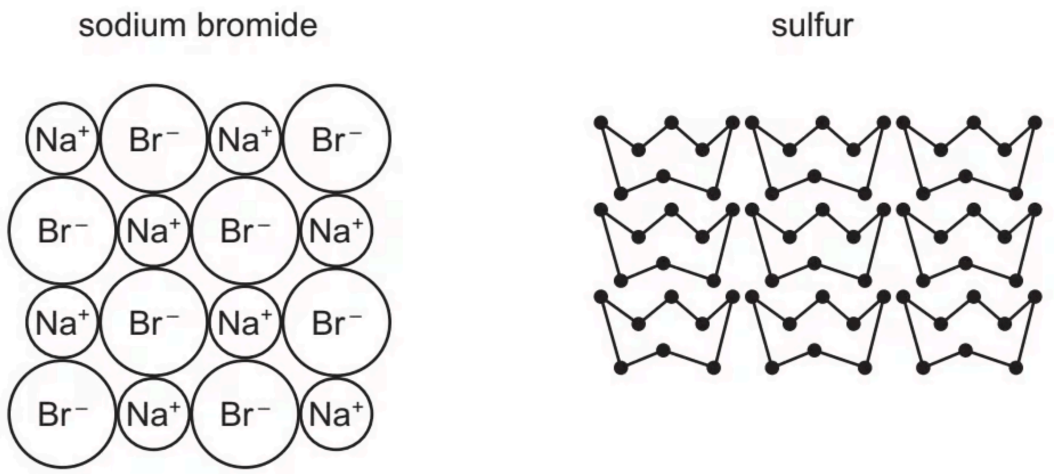
(5 marks)

(b) Complete the following table.

Gas	Test for Gas
Ammonia	
	Bleaches damp litmus paper
Hydrogen	
	Relights a glowing splint
	Turns limewater milky

(5 marks)

3 (a) The diagram shows part of the structures of sodium bromide and sulfur.



Describe both sodium bromide and sulfur in terms of:

bonding

electrical conductivity

solubility in water

....

....

....

....

....

(5 marks)

(b) Sulfur is an element.

What is meant by the term element?

....

(1 mark)

(c) A coloured crystal of cobalt(II) chloride is placed at the bottom of a beaker containing water. Colour spreads throughout the water over time. **Figure 1** shows the spread of colour after two days.

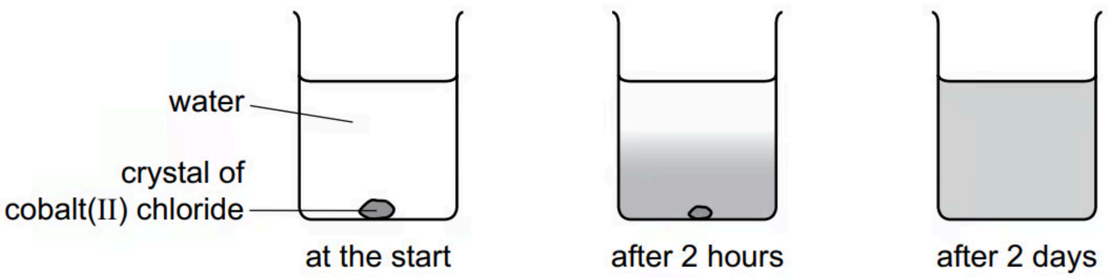


Figure 1

Cobalt is a transition element. Lithium is a Group I element.

Describe **two** ways in which the properties of cobalt differ from those of lithium.

....

....

(2 marks)

4 (a) A student investigates the reaction of small pieces of zinc with dilute sulfuric acid at 20 °C.
The zinc is in excess.
Sulfuric acid is a compound.

Define the term compound.

....

(1 mark)

(b) State the formula of the ion that is present in an aqueous solution of all acids.

....

(1 mark)

(c) A few drops of the indicator methyl orange are added to aqueous dilute sulfuric acid.
State the colour change observed

from orange to ____________________

....

(1 mark)

(d) The formula of sulfuric acid is H_2SO_4. Complete **Table 1** to calculate the relative molecular mass of sulfuric acid.

Table 1

atom	number of atoms	relative atomic mass	
hydrogen	2	1	$2 \times 1 = 2$
sulfur			
oxygen			

....

....

(2 marks)

5 (a) Iron is extracted from its ore, hematite, in a blast furnace.

Substances added to the furnace are:

- Iron ore, hematite, containing impurities such as silica, SiO_2
- Air
- Coke, C
- Limestone, $CaCO_3$

Substances formed in the blast furnace are:

- Molten iron
- Molten slag
- Waste gases such as carbon dioxide

State the **two** functions of the coke used in the blast furnace.

....

....

(2 marks)

(b) Write an equation for the conversion of hematite, Fe_2O_3, to iron.

....

....

(2 marks)

(c) Explain how the silica impurity is removed and separated from the molten iron.

....

....

....

(3 marks)

(d) The molten iron from the furnace is impure. It contains impurities which include the element carbon. Explain how the carbon is removed. Include an equation in your answer.

....

....

....

(3 marks)

6 (a) Give three differences in physical properties between the Group I metal, potassium, and the transition element, iron.

....

....

....

(3 marks)

(b) The following metals are in order of reactivity.

potassium
zinc
copper

For those metals which react with water or steam, name the products of the reaction, otherwise write 'no reaction'.

Potassium:
Zinc:
Copper:

....

....

....

....

....

(5 marks)

7 (a) Islay is an island off the west coast of Scotland. The main industry on the island is making ethanol from barley.

Barley contains the complex carbohydrate, starch. Enzymes catalyse the hydrolysis of starch to a solution of glucose.

Yeast cells are added to the aqueous glucose. Fermentation produces a solution containing up to 10% of ethanol.

i) Complete the word equation for the fermentation of glucose.

glucose → .... + ...

[1]

ii) Explain why is it necessary to add yeast and suggest why the amount of yeast in the mixture increases.

[2]

iii) Fermentation is carried out at 35°C. For many reactions a higher temperature would give a faster reaction. Why is a higher temperature not used in this process?

[2]

(5 marks)

(b) The organic waste, the residue of the barley and yeast, is disposed of through a pipeline into the sea. In the future this waste will be converted into biogas by the anaerobic respiration of bacteria. Biogas, which is mainly methane, will supply most of the island's energy.

i) Anaerobic means in the absence of oxygen. Suggest an explanation why oxygen must be absent.

[1]

ii) The obvious advantage of converting the waste into methane is economic.

Suggest **two** other advantages.

[2]

(3 marks)

8 (a) Molten potassium bromide can be electrolysed. Predict the products of this electrolysis at:

the anode the cathode

....

....

(2 marks)

(b) Electrolysis of concentrated aqueous sodium chloride using inert electrodes forms chlorine, hydrogen and sodium hydroxide.

What is meant by the term electrolysis?

....

....

(2 marks)

(c) Name a substance that can be used as the inert electrodes.

....

(1 mark)

9 (a) A student investigates the rate of reaction of small pieces of calcium carbonate with an excess of hydrochloric acid of concentration 1 mol/dm^3.

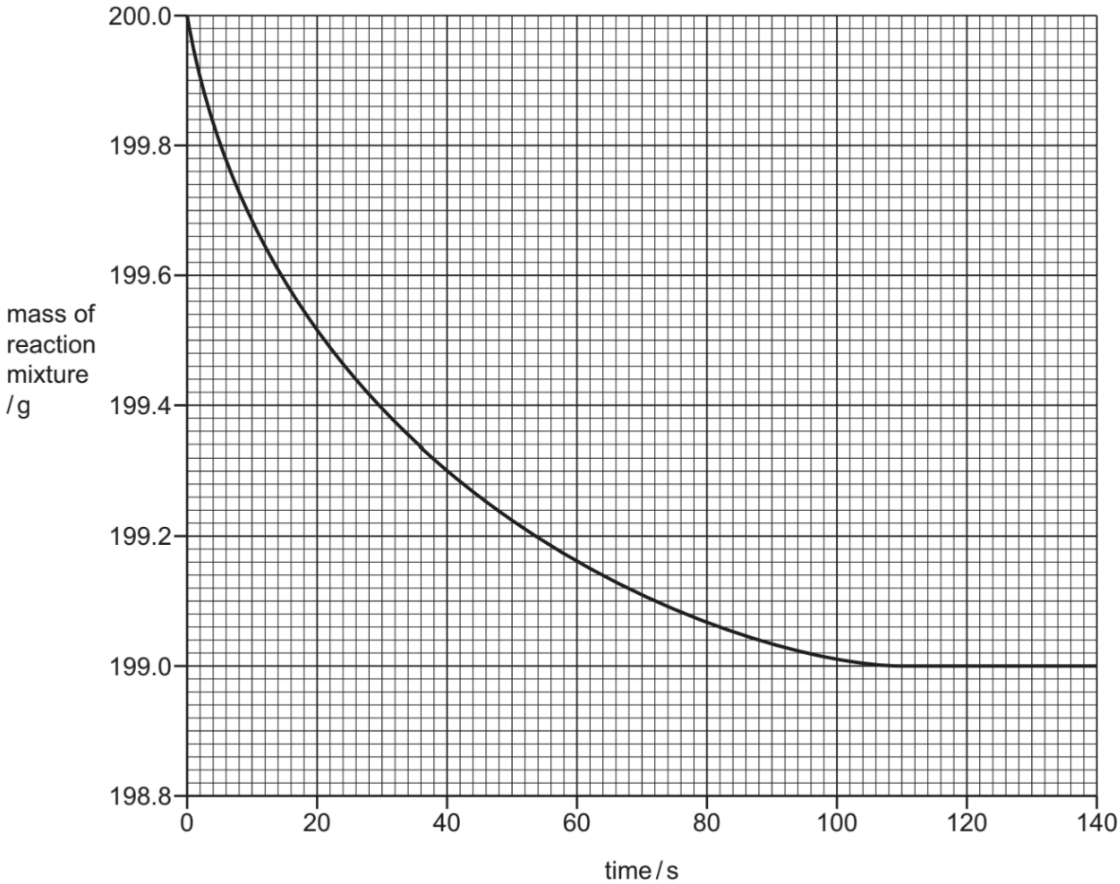
$$CaCO_3\,(s) + 2HCl\,(aq) \rightarrow CaCl_2\,(aq) + CO_2\,(g) + H_2O\,(l)$$

Name the salt formed when calcium carbonate reacts with hydrochloric acid.

....

(1 mark)

(b) The graph shows how the mass of the reaction mixture changes with time.



State why the reaction mixture decreases in mass.

....

(1 mark)

(c) Calculate the loss in mass during the first 40 seconds of the experiment.

(1 mark)

(d) The experiment is repeated using hydrochloric acid of concentration 2 mol/dm^3 .

All other conditions are kept the same.

Draw a line **on the grid** for the experiment using hydrochloric acid of concentration 2 mol/dm^3.

(2 marks)

(e) In the experiment, when 2.00 g of calcium carbonate is used, the loss in mass of the reaction mixture is 0.88 g. All other conditions are kept the same.

Calculate the loss in mass when 0.50 g of calcium carbonate is used.

(1 mark)

(f) The experiment is repeated using the same mass of different-sized pieces of calcium carbonate. All other conditions are kept the same. The sizes of the pieces of calcium carbonate are:

- powder
- small pieces
- large pieces.

Complete the table by writing the sizes of the pieces of calcium carbonate in the first column.

size of pieces of calcium carbonate	initial rate of loss in mass in g/ s
	0.005
	0.030
	0.100

(1 mark)

10 (a) Polluted air contains two oxides of carbon and two oxides of nitrogen. A major source of these pollutants is motor vehicles.

i) Describe how carbon dioxide and carbon monoxide are formed in motor vehicle engines.

[3]

ii) State one adverse effect of each of these gases.

[2]

iii) Nitrogen monoxide, NO, is released by motor vehicle exhausts. Explain how nitrogen monoxide is formed in motor vehicle engines.

[2]

iv) When nitrogen monoxide is released into the atmosphere, nitrogen dioxide, NO_2, is formed. Suggest an explanation why this happens.

[1]

(8 marks)

(b) Predict the possible adverse effect on the environment when this non-metal oxide, NO_2, reacts with water and oxygen.

(2 marks)